

SS2 Part A: Related Work and Data Description

Introduction:

This document presents a review of three related publications that address problems similar to this study's own concentration risk classification problem, followed by a description of the publicly available dataset selected for use in this study. As established in the project's motivation and problem statement, this study aims to investigate whether portfolio concentration measures can be used to predict elevated risk of a concentration-amplified loss. Since STADIOEquities' own client data is not yet available, SEC Form 13F institutional holdings data has been selected as a publicly available proxy dataset to validate this approach. The three publications reviewed below inform the modelling, preprocessing, and evaluation decisions made in this study, and at least one directly applies machine learning to the same public dataset used here.

Table 1 (Kacperczyk et al., mutual fund concentration paper):

Title	On the Industry Concentration of Actively Managed Equity Mutual Funds
Harvard reference	Kacperczyk, M., Sialm, C. and Zheng, L. (2005) 'On the Industry Concentration of Actively Managed Equity Mutual Funds', The Journal of Finance, 60(4), pp. 1983–2011.
Link	https://faculty.mcombs.utexas.edu/clemens.sialm/mfconcjf.pdf
Modelling	Uses a multivariate panel regression with panel-corrected standard errors (following Beck and Katz, 1995) to control for contemporaneous correlation, heteroskedasticity, and within-fund autocorrelation. The key explanatory variable is the Industry Concentration Index (ICI), a market-adjusted Herfindahl Index with a correlation of 0.93 to the standard Herfindahl Index. This is regressed against several performance measures: the Carhart (1997) four-factor abnormal return (both unconditional and conditional, following Ferson and Schadt, 1996), holding-based DGTW (1997) measures (CS and CT), industry-adjusted measures (IS and IT), and the Treynor-Black (1973) appraisal ratio. Control variables include expense ratio, turnover, log of age, log of total net

	assets, and new money growth, with time fixed effects included throughout. This is econometric panel regression, not a supervised machine learning classifier. No ensembling or regularisation is used.
Preprocessing Techniques	Three datasets were merged: the CRSP Survivorship-Bias-Free Mutual Fund Database, the CDA and Spectrum holdings database, and CRSP stock price data. These were matched manually by fund name and cross-verified using total net assets and investment objective where matching was ambiguous. Balanced, bond, index, international, and sector funds were excluded, retaining only actively managed domestic equity funds. Only the dominant share class was included for funds with multiple classes. Fund-quarters with fewer than 11 identifiable stock holdings were excluded, as were any fund-quarters where prior-quarter total net assets fell below \$1 million. Age and total net assets were log-transformed, since both are right-skewed. Explanatory variables were lagged by one quarter (expenses and turnover were lagged by one year) specifically to mitigate endogeneity. The final sample included 1,771 distinct funds from 1984 to 1999.
Datasets used	The CRSP Survivorship-Bias-Free Mutual Fund Database, the CDA and Spectrum mutual fund holdings database, the CRSP stock price database, and Kenneth French's 48-industry SIC classification, aggregated into 10 broader industry groups.
Key takeaways	1. The concentration measure used is explicitly a market-adjusted Herfindahl Index, empirically almost identical to the plain Herfindahl Index, with a correlation of 0.93. This directly supports the use of HHI as a defensible, literature-backed concentration feature for this study. 2. The central finding is that higher concentration is associated with better performance on average, driven by genuine stock-picking skill. This is an important nuance, showing that concentration is not simply equivalent to risk being bad. It directly justifies why this study's label deliberately targets the downside tail, meaning concentration-amplified loss, rather than average outcomes. This is a meaningful and explainable departure from the average-outcome framing used in this paper. 3. The deliberate use of one-quarter-lagged explanatory variables to avoid endogeneity directly reinforces this study's own filing-date-based information cut-off, applying the same lagging logic to avoid label leakage.

Table 2 (Yang et al., portfolio optimisation paper):

Title	Portfolio Optimization with Idiosyncratic and Systemic Risks for Financial Networks
Harvard reference	Yang, Y., Zhao, L., Chen, L., Wang, C. and Han, J. (2021) Portfolio Optimization with Idiosyncratic and Systemic Risks for Financial Networks. arXiv preprint.
Link	https://arxiv.org/abs/2111.11286
Modelling	Proposes a multi-objective convex portfolio optimization model minimising a weighted combination (via trade-off parameter α) of the variance-covariance matrix (idiosyncratic risk) and a network-derived interconnectedness matrix built from each asset's average clustering coefficient within a correlation-based network (systemic risk). Three correlation network types are tested: Pearson, Kendall's tau, and lower-tail dependence. Solved via the CVX convex-optimization toolkit under a rolling-window scheme (12-month in-sample, 1-month out-of-sample, stepped monthly, 75 windows total). Benchmarked against the classic Markowitz mean-variance model (MV) and Clemente et al.'s (2021) clustering-only model (CGP). This is portfolio optimization, not supervised ML, no ensembling/regularisation in the ML sense.
Preprocessing Techniques	Uses raw daily log returns (no explicit missing-value handling or feature scaling reported, since inputs are direct return series, not engineered tabular features). Rolling-window resampling (12-month windows, monthly step) is the core "preprocessing" mechanism, used to re-estimate the correlation networks and covariance matrices at each step before re-optimizing.
Datasets used	Two Chinese equity datasets sourced from Joint Quant: SSE 180 index constituents (126 stocks) and SSE 50 index constituents (42 stocks), daily returns from April 2014 to August 2021.
Key takeaways	1. The modified Herfindahl Index (HI_w) is used purely as a post-hoc diversification metric to evaluate optimised portfolios, not as an input to the optimisation objective. This confirms that HHI-style measures are a standard, literature-accepted way to quantify concentration after the fact, directly supporting this study's own use of HHI as a feature. 2. The paper explicitly pairs HHI alongside drawdown as complementary evaluation metrics, reporting both together

	in Table 2 and Table 4. This is genuine literature support for evaluating concentration and downside loss jointly, matching this study's own portfolio-level, concentration-weighted loss label design. 3. The finding that a lower alpha, meaning less weight placed on systemic or network risk, tends to produce more diversified portfolios with a lower HI_w, reinforces the intuitive, literature-grounded relationship between a portfolio's risk-management approach and its resulting concentration level. This supports the general premise that concentration is a meaningful, controllable risk dimension worth modelling.
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Table 3 (Chen et al., SEC 13F ML paper):

Title	Applying Machine Learning to SEC 13F Investment Manager Filings for Portfolio Construction and Rebalancing
Harvard reference	Chen, Y., Song, Y., Qian, S. and Fleiss, A. (2023) Applying Machine Learning to SEC 13F Investment Manager Filings for Portfolio Construction and Rebalancing. SSRN Working Paper.
Link	https://ssrn.com/abstract=4352145
Modelling	Trains three tree-ensemble models, Random Forest, XGBoost, and CatBoost, to predict Return ₂ , defined as the security's subsequent-quarter return, using engineered 13F-derived factors. Predictions are used to sort securities into 10 quantile-based portfolios per model, which are then compared against the S&P 500 benchmark and the average of all institutional investors' actual portfolios. CatBoost additionally incorporates a categorical industry feature, taking advantage of its native categorical-feature handling. This was the only architectural difference between the three models tested. No explicit regularisation or hyperparameter tuning details are reported. Ensembling is inherent to all three algorithms: bagging for Random Forest, boosting for XGBoost and CatBoost.
Preprocessing Techniques	Two raw sources: historical SEC 13F filings (2013–2020) and daily OHLC stock price data, each cleaned separately before merging. Price data: cleaned abnormal/missing values, fixed inconsistent measurement units, imputed missing prices using Yahoo Finance and Bloomberg. 13F data: dropped duplicates and missing values,

	then filtered out records with low market value (<100K) or outstanding shares (<10K), and excluded records filed before 2013/06/30 (before which 13F filing wasn't mandatory). Industry classifications were joined in via Yahoo Finance. The two cleaned tables were merged on CUSIP and PERIOD_END.
Datasets used	Historical SEC Form 13F filings (2013–2020), scraped directly from SEC.gov, using a schema with fields: RECORD_ID, CIK, CUSIP, PERIOD_END, FILING_DATE, QTY, MARKET_VALUE; and a separate daily stock price dataset with CUSIP, EXCHANGE, TICKER, DATE, VOLUME, OPEN, HIGH, LOW, CLOSE.
Key takeaways	1. This paper's 13F schema explicitly separates PERIOD_END from FILING_DATE as two distinct fields, confirming that a complete 13F dataset can support proper, leakage-free labelling. This is directly useful for this project, since it shows that even if the Kaggle dataset used here lacks a separate filing date field, the correct underlying schema does include one. This reinforces that a 45-day-lag approximation is a reasonable substitute for a genuinely missing field, rather than a shortcut around available information. 2. The paper explicitly avoids look-ahead bias by using Return2, the subsequent-quarter return, as the label, specifically because the exact positions held by institutions are unknown before the 13F form is filed for each quarter. This is the same reasoning this project uses to justify predicting a future-period outcome from a filing-date-lagged snapshot, giving a direct, citable precedent for that methodological choice. 3. The finding that the strategy is not market neutral, and can generate alpha in bullish markets but is inconsistent in bearish markets, with large drawdowns during the March 2020 crash, is a useful caution. Models built on 13F data may behave very differently across market regimes, which is worth acknowledging as a limitation of using a fixed 2015 to 2017 window to train and evaluate this project's own concentration-risk model.

Table 4 (the dataset table):

Name	SEC 13F-HR Institutional Investment Data
Harvard reference	Panoli, A. (2020) SEC 13F-HR Institutional Investment Data. Kaggle.
Link	https://www.kaggle.com/datasets/aneeshpanoli/sec-13fhr-institutional-investment-data
Number Instances Of	1,713,673 holding-level records across 2,222 unique institutions and 41,244 unique securities, spanning 37 monthly snapshot columns (2015-01 to 2017-09), of which 11 correspond to actual 13F quarter-end reporting periods (March, June, September, December). Two supplementary lookup tables are also provided: institutions.csv (2,222 rows, mapping institution CIK to name) and stock_names.csv (41,244 rows, mapping CUSIP to security name).
Number feature columns of	4 identifier columns (cusip, stock, cik, institution), plus 33 monthly share-count columns (2015-01 to 2017-09), giving 37 columns in total in the main file.
Number classes of	Not applicable in the raw dataset, since this is holdings data, not pre-labelled. The classification target (concentration-amplified loss risk) is constructed as part of this study's own feature and label engineering and is not present natively.
Target range	Not applicable. A regression target is not used. A binary classification label is engineered instead, as described in the problem statement.
Origin	Extracted from SEC Form 13F-HR filings submitted to the SEC's EDGAR system by institutional investment managers with \$100 million or more in qualifying U.S. equity assets under management, who are legally required to file quarterly. Values represent number of shares held. The dataset does not include dollar values, tickers, or sector classifications, all of which are sourced separately as part of this study's preprocessing.